AAC Access Systems

User Manual for Setup, Practice, and Testing

Based on AAC Access Systems v0.55

Purpose: help AAC students compare access systems by holding the task constant while varying input device, selection technique, selection set/layout, and specialized AAC interface design.

1. Overview

AAC Access Systems is a classroom demonstration and practice app for examining how different AAC access systems change the work required to produce the same message. The app is not a clinical prescription tool. It is a teaching and analysis tool for making access-system architecture visible: what the user acts with, what choices are available, how the choices are selected, and how efficiently the target message can be produced.

Core concepts

Category	Meaning in this app	Examples
Input device	The device or control channel by which the user controls the target system.	Touchscreen, mouse/trackball, physical keyboard, switch, joystick, head tracker.
Selection set / layout	The visual, auditory, or tactile set of available items from which the user chooses.	Alphabetic keyboard, QWERTY keyboard, frequency keyboard, 4x4/5x5/6x6 colored-shape board, Morse dot/dash set, prediction list, stored phrase page.
Selection technique	The means by which the user chooses an item from the selection set and sends it to the system.	Direct selection, dwell, single-switch scanning, row-column scanning, group-item scanning, two-switch step scanning, joystick navigation, head tracking plus dwell, Morse encoding.
Task / information set	The message or sequence the user is trying to produce.	A text sentence such as “I need help with my wheelchair” or a five-item colored-shape sequence.

2. Screen areas

Screen area	What it is used for
AAC Access Systems setup pane	Choose the task type, edit or generate the target, and select which systems should be available for comparison.
Task and information set	Choose Text sentence or Colored-shape sentence. For text, type or load a preset. For shapes, choose the board size and generate a five-item target.
Input Device / Selection Technique	Choose access methods such as direct touch/click, mouse/trackball, scanning, joystick, or head tracking.
Selection Set / Layouts: keyboard	Choose text keyboard organizations: alphabetic, QWERTY, or frequency.
Selection Set / Layouts: shapes	Choose colored-shape boards: 4x4, 5x5, or 6x6.
Selection Set / Layouts: specialized	Choose specialized systems such as Morse, word

	prediction, abbreviation expansion, core vocabulary, stored phrase, or icon sequence.
User interface / student production	The main hands-on workspace. Students practice, test, and simulate the selected interface here.
Maximum-efficiency simulation	Runs the model's optimal path and displays selection-efficiency statistics and event logs.

Important layout note. In the student production workspace, the Test field, Output field, Active interface line, and run controls remain visible while the user works in the interface. The Help button stores interface-specific instructions that would otherwise take up space above the keyboard.

3. Quick-start workflow

1. Open the AAC Access Systems setup pane.
2. Choose the task type: Text sentence or Colored-shape sentence.
3. Set the target. For text, use a preset or type a sentence. For shapes, choose the board size and generate a five-square target.
4. Select the Input Device / Selection Technique choices you want students to compare.
5. Select the Selection Set / Layout choices: keyboard layouts, shape layouts, and/or specialized systems.
6. Open User interface / student production.
7. Choose the student interface using the Selection technique, Keyboard layout, and Shapes / specialized layouts menus.
8. Configure timing or access options, such as dwell time, keypress delay, scan delay, Morse boundaries, or head-tracking settings.
9. Use Practice so the student can explore the interface without scoring.
10. Use Test when the student is ready to produce the target under the selected interface.
11. Review Student analysis and the Student event log.
12. Use Simulate to compare the student run with the maximum-efficiency path for the same interface.

4. Setting up the task

Text sentence mode

13. In Task and information set, choose Text sentence.
14. Type a sentence in the text box or select a preset and press Load preset.
15. Keep the target short enough for the instructional goal. A short sentence is useful for introducing a new selection method; a longer sentence is useful for comparing efficiency across systems.
16. Avoid unsupported characters when possible. The app is primarily organized around lowercase letters, space, punctuation, and selected symbols.

Colored-shape sentence mode

17. Choose Colored-shape sentence.
18. Choose a 4x4, 5x5, or 6x6 colored-shape board.
19. Press Generate random 5-square sentence.
20. Confirm that the corresponding shape layout is selected in Selection Set / Layouts: shapes.
21. Use this mode when the instructional focus is machine access, target acquisition, scanning depth, or selection-set organization without language formulation demands.

Teaching note. The colored-shape task is useful when you want students to focus on access demands rather than vocabulary, spelling, or prediction. The target is still a sequence, but it removes much of the linguistic content of the task.

5. Building the system set

The setup pane allows you to create the set of systems that will be available for student practice, testing, and simulation. A complete ordinary interface is created by crossing a selection technique with a layout. Specialized interfaces appear as complete systems.

Setup choice	How to use it
Core set	Selects a practical default set for classroom comparison. Use this when first introducing the app.
All	Selects every available technique, layout, and specialized system. Use this for broad demonstrations, not for first-time student practice.
None	Clears all choices so you can build a focused comparison set.
Technique checkboxes	Choose how the user will select items, such as direct click, row-column scanning, or head tracking.
Keyboard layout checkboxes	Choose the text selection sets that will be crossed with the selected techniques.
Shape layout checkboxes	Choose colored-shape selection sets for shape-based demonstrations.
Specialized layout checkboxes	Choose systems that include their own representation or encoding scheme, such as Morse, abbreviation expansion, or stored phrases.

6. Choosing the active student interface

In User interface / student production, the active interface is controlled by three menus. These menus let the instructor separate the access method from the layout and from specialized systems.

Menu	Use
Selection technique	Choose the access method, such as direct touch/click, mouse/trackball, scanning, joystick, or head tracking.
Keyboard layout	Choose the text keyboard arrangement paired with the selected technique.
Shapes / specialized layouts	Choose a colored-shape board or a specialized AAC system. This menu overrides the ordinary keyboard layout when a shape or specialized system is selected.
Interface width	Choose how much horizontal screen width the interface occupies. Morse defaults to 0.5 width; other interfaces may use full width or narrower widths depending on the lesson.

Active interface line. The Active interface field reports the currently selected system. The Highlight on/off button next to it controls whether the Test field marks completed text and the next target item.

7. Configuring access options

Option area	Relevant systems	What to set

Direct / pointer selection options	Direct touch/click and mouse/trackball	Full screen, dwell selection, dwell time, and keypress delay. Keypress delay is a hold-to-accept threshold.
Scanning options	Single-switch and step-scanning systems	Scan delay and selection delay. These determine the pace of the scanning demonstration and the modeled activation point.
Prediction / specialized options	Word prediction and character prediction systems	Prefix length and character prediction list size.
Morse code options	Morse code input	Keyboard mapping and timing boundaries. Dot/dash input can use period/hyphen or left/right arrows. Letter and word boundaries can be adjusted.
Head tracking controls	Head tracking plus dwell or switch	Start camera, test camera, calibrate center, sensitivity, dead zone, smoothing, and head dwell time. Camera access depends on browser permission and context.

Full screen. Use Full screen when the student needs a larger workspace or when the instructor is projecting the interface. Full screen is placed with the direct/dwell controls because it is most often needed when students are actively interacting with the interface.

Highlight on/off. When Highlight is on, the Test field marks the portion already produced and the next item. When Highlight is off, the Test field shows the target without progress marking. This is useful when the visual cue is helpful for demonstration but would provide too much support during testing.

8. Practice mode

Practice is used before testing. It lets the student explore how the interface works without scoring target mismatch errors.

22. Choose the active student interface.
23. Configure any relevant timing or access options.
24. Press Practice.
25. Ask the student to try several selections before attempting the target. For scanning, have the student watch at least one full scan cycle. For Morse, have the student try several dot/dash sequences. For shape mode, have the student identify a target shape and then select it.
26. Use the Help button if the student needs the current interface instructions.
27. Use Reset to clear the practice output and begin again.

Practice observation	What to look for
Can the student identify the active selection set?	Do they know where the available choices are and how the layout is organized?
Can the student predict what the next action will do?	For scanning, do they understand row vs item phase? For step scanning, do they understand advance vs select?
Are access timing settings appropriate?	Dwell, keypress delay, scan delay, and Morse boundaries should be long enough for success but short enough to preserve task flow.
Does the student use Help appropriately?	Help should support orientation without occupying the main interface space.
Does the display remain visible?	Test, Output, controls, and active interface information should stay visible during the task.

9. Test mode

Test mode is used when the student is ready to produce the target through the selected interface. The app records selections, activations, scan ticks, navigation actions, prediction selections, code elements, corrections, errors, and output.

28. Confirm that the target is correct.
29. Confirm that the desired interface is shown in the Active interface field.
30. Decide whether Highlight should be on or off for the test.
31. Press Test.
32. Have the student produce the target using only the active interface.
33. Press Speak / complete when the produced output should be treated as complete, or allow the run to finish when the target is completed.
34. Review Student analysis and Student event log.
35. Use Reset before starting a new test run.

Instructor decision point. If the goal is to teach the mechanics of the interface, leave Highlight on. If the goal is to evaluate independent use of the interface, turn Highlight off after the student understands the task.

10. Using Simulate and Student analysis

Simulate plays the maximum-efficiency path for the active interface. It is useful for comparing what the system makes possible with what the student actually did.

36. Choose the same interface used for the student run.
37. Press Simulate from the student production controls, or use the Maximum-efficiency simulation pane.
38. Watch the interface animate the optimal path.
39. Compare the student run with the simulation in Student analysis.
40. Use the event logs to identify where extra selections, scan ticks, or navigation actions occurred.

Measure	Interpretation
Selections	How many chosen items were required to produce the target.
Activations	How many physical or simulated input acts were required.
Scan ticks	How many scanning advances occurred before selections. High scan ticks may indicate waiting cost.
Navigation actions	Moves between rows, groups, pages, or focus positions.
Prediction selections	Selections from prediction lists.
Code elements	Elements in an encoding scheme, such as dot/dash components in Morse.
Selections per character / word	A compact index of production efficiency for the selected system.
Characters per selection	How much output information is produced per selection. Higher values often reflect prediction, stored phrases, or encoding systems.

11. Special workflows

Morse code input

41. Choose Morse code input from Shapes / specialized layouts.
42. Set the Morse key mapping: period/hyphen or left/right arrows.
43. Set letter and word boundary durations.
44. Use Practice first. Have the student enter simple letters such as e, t, a, and n before attempting a sentence.
45. Use the Morse guide beside the input controls to support early practice.
46. Move to Test when the student can reliably produce letters and word spaces under the selected timing boundaries.

Colored-shape layouts

47. Choose Colored-shape sentence in the task pane.
48. Generate a five-square target.
49. Choose the matching shape layout in Shapes / specialized layouts.
50. Use Practice to teach the student how shape/color combinations are organized.
51. Use Test to measure access performance without spelling or vocabulary demands.

Head tracking

52. Choose Head tracking + dwell or Head tracking + switch as the selection technique.
53. Open Head tracking controls.
54. Use Test camera if permission is uncertain.
55. Start camera and then Calibrate center when the user is positioned comfortably.
56. Adjust sensitivity, dead zone, smoothing, and dwell time.
57. Practice aiming before starting a test run.

Camera note. Webcam access depends on the browser and how the HTML file is opened. If the camera does not start, try opening the file in its own browser tab or using a secure/localhost context, then allow camera permission.

12. Case study: SLP graduate student in an AAC course

Case: Maya is a first-year speech-language pathology graduate student enrolled in an AAC course. She has learned the definitions of input device, selection set, and selection technique, but she tends to describe AAC systems only by the visible keyboard. The instructor uses AAC Access Systems to help her separate the physical control method from the selection set and from the selection technique.

Teaching goal

- Maya will set up two systems with the same target sentence and explain how the access demands differ.
- Maya will complete a practice run before testing.
- Maya will interpret selections, activations, scan ticks, and event logs after the test.

Setup

58. The instructor chooses Text sentence and loads: “I need help with my wheelchair.”
59. In Input Device / Selection Technique, the instructor selects Direct touch/click and Single-switch row-column scan.
60. In Selection Set / Layouts: keyboard, the instructor selects Alphabetic 6x6.
61. In User interface / student production, Maya first chooses Direct touch/click with Alphabetic 6x6.
62. The instructor leaves Highlight on for the first practice run.

Practice phase

Maya presses Practice and selects several letters and spaces. She uses Help once to confirm that selecting a key enters that character. The instructor asks her to name the components: the input device is touch/click, the selection set is the alphabetic keyboard, and the selection technique is direct selection.

Test phase 1: direct selection

63. Maya presses Reset and then Test.
64. She produces the target sentence using the alphabetic keyboard.
65. She presses Speak / complete when the sentence is finished.
66. She reviews the Student analysis panel and notes that the number of selections is close to the number of characters and spaces produced.

Test phase 2: row-column scanning

67. Maya changes the Selection technique to Single-switch row-column scan while keeping the alphabetic keyboard layout.
68. She practices one full row scan and one item scan before testing.
69. The instructor turns Highlight off to reduce cueing.
70. Maya presses Test and produces the same target sentence.
71. In the Student analysis panel, Maya compares the scanning run with the direct-selection run and notices that scanning added scan ticks and additional selections for row and item choices.

Debrief questions

- What stayed the same across the two tests? The target sentence and the alphabetic selection set.
- What changed? The selection technique changed from direct selection to row-column scanning.

- Why did the scanning condition require more waiting? The system had to cycle through rows and then items before each character could be selected.
- How would this matter clinically? A client may be able to use the same vocabulary layout but need a different selection technique because of motor access constraints. The interface may look similar while the access demands are very different.

Outcome. By the end of the activity, Maya can explain that “alphabetic keyboard” names the selection set, not the whole access system. She can also explain why row-column scanning changes the user’s workload even when the visible keyboard remains the same.

13. Instructor checklist

Before class	During practice	During test	After test
Choose target mode and sentence/shape sequence.	Use Practice first; let students explore without scoring.	Confirm active interface and highlight setting.	Open Student analysis and compare with Simulate.
Select a focused system set rather than every available system.	Prompt students to name input device, selection set, and selection technique.	Use Reset before each run.	Review event logs for extra selections, scan ticks, or navigation actions.
Check display size and use Full screen if needed.	Use Help only when needed so the main interface remains uncluttered.	Press Speak / complete when the output should count as complete.	Ask students to interpret why systems differed, not only which was faster or shorter.

14. Troubleshooting

Problem	Likely cause	What to try
The shape-layout checkboxes do not appear relevant in text mode.	Shape layouts are most meaningful with the colored-shape task.	Use Colored-shape sentence or select a shape layout as the active specialized layout for demonstration.
The output field clips shape tokens.	The output field has a fixed compact height.	Use the current compact output tokens; scroll horizontally if the sequence exceeds the field width.
The target highlight is too helpful during testing.	Highlight is turned on.	Use the Highlight on/off button beside Active interface.
The student does not know what the interface expects.	The instructions are stored in Help to save screen space.	Press Help under Speak / complete.
Scanning is too fast or too slow.	Scan delay and selection delay need adjustment.	Change Scanning options before Practice or Test.
Morse letters complete too soon or too late.	Letter or word boundary durations do not fit the student’s rhythm.	Adjust Morse code options and practice simple letters again.
Head tracking camera fails.	Browser permission, embedded context, or insecure context.	Use Test camera, open the file in its own tab, allow camera permission, or try a secure/localhost context.
The interface is too small.	Interface width or browser display is constrained.	Use Full screen, select a wider interface width, or reduce the browser zoom level.

Use in instruction. For AAC coursework, the most important outcome is not simply identifying which system has the fewest selections. The goal is to help students articulate how access method, selection set, selection technique, and encoding/prediction architecture jointly shape the work of message production.